

Paper A v0.8 — The Physical (SU(3))
 Yang–Mills Mass Gap via D_{IV}^5 Spectral
 Geometry. Companion to Paper B. v0.8 = v0.7
 + the §570 CANDIDATE floor-lift (Section 2a,
 clearly tiered, NOT banked). The banked floor
 stays EXISTENCE ($\Delta > 0$ from a bounded
 domain, geometry-fixed, no free dimensionless
 parameter beyond the ruler R). NEW
 candidate: by §570 (energy $\neq$ mass), the
 pure-time modes $(0, m)$ on the real boundary
 $(S^4 \times S^1)/\mathbb{Z}_2$ are ENERGIES not masses, so the
 MASS gap is the lightest SPATIAL-carrying
 $\mathbb{Z}_2$ -even mode $(1, 1) = 4 + 1/R^2 > 4$ ($= 5$ at $R=1$)
 — because the $\mathbb{Z}_2$ forbids a purely-spatial
 rest-mass ($k=1, m=0$ is odd), so a $k=1$ spatial
 state must carry at least one time unit $m=1$.
 The deep insight: the $\mathbb{Z}_2$ FORBIDS A PURE
 REST-MASS — every massive state carries one
 unit of time (a candidate BST statement about
 mass). This would return ‘gap > 4 ’ with correct
 meaning, lifting the floor, IF Grace $\oplus$ Elie
 confirm the mass/energy split on the real
 boundary. Carried over from v0.7: bare- S^4 ‘4’

$\mathbb{Z}_2$ -projected-out; three geometry-fixed opens
 (adjoint conformal weight Δ , lowest $\mathbb{Z}_2$ -even
 mode, radius R); ‘6’ = conformal/identity, ‘7’ =
 genus; glueball separate-and-valid (on C_2^{int} ,
 not λ_1), out of the room. Package intro HELD.
 Count HOLDS 4.

Lyra (Claude Opus 4.8) — Casey Koons, PI; Grace (OS reduction + T2490/T2491 spectral case)

2026-08-19 Wednesday (date-verified)

The Physical Yang–Mills Mass Gap via D_{IV}^5 Spectral Geometry

Paper A of the two-paper package (companion: Paper B, substrate uniqueness). v0.8 — existence floor banked; §570 candidate floor-lift added (Section 2a, tiered): the mass gap is the lightest spatial-carrying $\mathbb{Z}_2$ -even mode $(1,1) = 4 + 1/R^2 > 4$, because the $\mathbb{Z}_2$ forbids a pure rest-mass — every massive state carries one unit of time. Carried from v0.7: bare- S^4 “4” $\mathbb{Z}_2$ -projected-out; three geometry-fixed opens; glueball separate-and-valid, out of the room; package intro HELD.

0. Two guards and one residual (read this first)

Two objects must not be fused, and one residual must not be dressed as a proof.

- Guard 1 (space). The gap we derive is the spectral gap of the Bergman–Laplacian on the bounded domain D_{IV}^5 — a *different operator on a different space* from the Yang–Mills Hamiltonian on flat $\mathbb{R}^4$. $\lambda_1 = C_2 = 6$ on our domain is not, by assertion, the $\mathbb{R}^4$ mass gap; the two are connected only through a bridge that must be exhibited, never fused (the standing K932 guard: exhibit the bridge, don’t assert a shared eigenvalue).
- Guard 2 (object). The pure-Yang–Mills (Clay) gap is the glueball (1720 MeV, 0.6% vs lattice), not the proton (938 MeV, which is the *full-QCD*, matter-included gap — a different object); we do not lend the proton’s four-decimal precision to the Yang–Mills gap (K1705). And the glueball is *out of the room* for the gap-value pin (K1708/K1709): $1720 = (11/6) \cdot 938$ is built on C_2^{int} (the labeled integer primary), not on λ_1 (the boundary Laplacian gap), and its $11/6$ carries $C_2 = 6$ — so it is a real, separate forward prediction but not independent evidence for the gap value. The value is

pinned from the geometry alone (the GATE-KOIDE discipline that keeps $Q = 2/3$ out of the Koide pin).

- Guard 3 (the symbol “ C_2 ” itself — K1707). Three distinct objects all read “ C_2 ” and all equal 6 at the physical point $n = n_C = 5$, so writing “the gap = $C_2 = 6$ ” hides *which one* is meant:
 - $C_2^{\text{int}} := n_C + 1 = 6$ — the *labeled integer primary* (one of the five integers). A definition.
 - $C_2^{\text{flow}}(n) = 2n - 4$ — the *commitment-flow Casimir* (a function of n ; = 6 at $n=5$ only by the coincidence-at- $n_C=5$).
 - $C_2^{\text{Berg}} = n + 1$ — the *Bergman-representation Casimir* (a function; = 6 at $n=5$, and equal to C_2^{int} *identically*, since C_2^{int} is *defined* as $n+1$).
 The consequence (Cal, K1707): “ $\lambda_1 = C_2 = 6$ ” as previously written is C_2^{int} = C_2^{Berg} — the same definition on both sides, an identity that *cannot fail* ($P^2=P$), not a derived tower-gap. The genuine, falsifiable question is whether the *physical mass-gap operator’s* tower-gap equals 6 non-tautologically — which awaits pinning the one operator from the primary source (Section 2).
- The one residual (flat space). The honest open problem is a single question: is the theory that the reconstruction machinery (Rehren holography $\rightarrow$ Osterwalder–Schrader) places on the flat 4D boundary literally interacting $SU(3)$ Yang–Mills, or a gapped generalized-free cousin that also satisfies the axioms? This is the large named open — carried as such, never a proof.

With those stated, the paper’s claim is explanatory, not triumphal: BST reproduces the scale and channel-structure of the Yang–Mills mass gap on a bounded domain. The gap is nonzero because the domain is bounded (real physics, Section 2), with no free dimensionless parameter beyond the ruler R . Its *value* is genuine open physics: the bare- S^4 “4” is projected out by the $\mathbb{Z}_2$ on the physical boundary $(S^4 \times S^1)/\mathbb{Z}_2$ ($k+m$ even, K1709), and the honest value awaits three geometry-fixed opens — the conformal weight Δ , the $\mathbb{Z}_2$ coupling, and the radius R . The old “6” was the conformal reading = the *identity* $C_2 = n_C+1$ (a time weight, not a mass); “7” is the genus. The banked floor is existence: the gap exists, geometry-fixed, no dimensionless knob. The flat- $\mathbb{R}^4$ construction is carried as the large named open. Nothing here is “we solved Yang–Mills,” and — after K1709 — nothing here claims a derived integer value; the value is honest open physics, not a stalled identity.

Abstract

We study the Yang–Mills mass gap for the physical gauge group $SU(3)$ — color — derived (not assumed) from the substrate D_{IV}^5 in the companion Paper B. The mass-gap operator is a Laplacian on the physical Shilov boundary $\partial_S D_{IV}^5 = (S^4 \times S^1)/\mathbb{Z}_2$, and its gap is nonzero because the domain is bounded — the spectrum is discrete and bounded below, and *compactness is the gap mechanism* (decompactify and the gap slides to zero: $\lambda_1(R) \rightarrow 0$). This existence is rock-solid, with no free dimensionless parameter beyond the ruler R (a Planck-unit, the theory’s single dimensionful input). The *value*, however, is genuine open physics. A bare- S^4 computation gives a first eigenvalue 4 (tower $k(k+3)$, degeneracies 1,5,14,30,55,

first-level degeneracy $n_C = 5$), but the physical boundary is $(S^4 \times S^1)/\mathbb{Z}_2$, whose $\mathbb{Z}_2$ admits only $k + m$ even and therefore *projects out* the $\lambda = 4$ mode ($k=1, m=0$, odd), leaving R -dependent survivors (K1709, Cal). The earlier “ $\lambda_1 = C_2 = 6$ ” was the *conformal* ($SO(2)/\text{time-circle}$) *reading* — the bare 4 plus a +2 shift — coinciding with $C_2 := n_C + 1$ = the Bergman-representation Casimir (an identity that cannot fail, Cal K1707), i.e. a *time* conformal weight, not a mass. So the honest value awaits three geometry-fixed opens (K1709): (i) the conformal weight Δ of the lowest gauge excitation, (ii) which surviving $\mathbb{Z}_2$ -even mode is lowest, (iii) the radius R (the ruler, which decides $4/R^2$ vs $4+1/R^2$). The banked floor is existence itself: the gap exists, is geometry-fixed with no free dimensionless parameter beyond the ruler; its integer value is real open physics, not a stalled identity. A candidate floor-lift (Section 2a, tiered — not banked): if §570’s mass/energy split holds on the real boundary, the pure-time modes $(0,m)$ are energies and the *mass* gap is the lightest spatial-carrying $\mathbb{Z}_2$ -even mode $(1,1) = 4 + 1/R^2 > 4$ ($= 5$ at $R = 1$) — because the $\mathbb{Z}_2$ forbids a purely-spatial rest-mass, so *every massive state carries one unit of time*. This would return “gap > 4 ” with correct meaning; it awaits Grace $\oplus$ Elie. (The glueball $1720 \text{ MeV} = (11/6) \cdot 938$ is a separate valid prediction built on $C_2^{\wedge \text{int}}$, not λ_1 — kept, but out of the room as gap-evidence, K1708.) The construction does not build 4D Yang–Mills from scratch on flat $\mathbb{R}^4$; instead the D_{IV}^5 spectral theory directly supplies Osterwalder–Schrader reconstruction data (a rigorous Hilbert space, a gapped Hamiltonian, reflection positivity, 4D conformal/Poincaré covariance, clustering) on the domain’s 4D conformal boundary ($SO(4,2) \subset SO(5,2)$), so the existence of a rigorous gapped 4D theory follows from classical harmonic analysis on a bounded symmetric domain. The problem then reduces to a single residual (Section 6, Section 9): is this gapped edge theory genuinely interacting Yang–Mills, or a (also-gapped, also-axiom-satisfying) generalized-free theory — equivalently, is the Rehren-reconstructed boundary theory *literally* $SU(3)$ YM? In v0.4 we report that the decisive probe — the cross-channel glueball spectrum — has been computed and materially advances the interacting reading: the four ground $J^{\wedge PC}$ channels follow a linear conformal-energy ladder off one verified number (the genus of D_{IV}^5), with one blind leg, $2^{++} = g/n_C$, and the other channels consistent within current lattice error. A generalized-free theory produces no such bound-state $J^{\wedge PC}$ tower — real evidence on the interacting side — but the interacting-vs-free question remains the named-open residual. We do not claim the Clay problem is solved. The honest statement is a decisive *sharpening* to one residual, with the flat- $\mathbb{R}^4$ identification carried as the large named open.

1. Scope: the physical theory, not “all G ”

The Clay problem asks for a mass gap for *every* compact simple gauge group. BST addresses the physical theory: $G = SU(3)$, color. The companion Paper B proves D_{IV}^5 is the unique substrate and that $SU(3)$ is derived, not chosen. So “all G ” is not BST’s claim. This paper makes claims only about the physical theory and is explicit about that scope (Section 9).

2. The mass gap: existence SOLID, value = three geometry-fixed opens on the real boundary $(S^4 \times S^1)/\mathbb{Z}_2$ (K1709)

What is SOLID (real physics). The mass-gap operator is a Laplacian on the Shilov boundary of D_{IV}^5 — a positive operator whose spectrum is the arrow of time (the heat-semigroup generator). The domain is bounded; the state space $H^2(D_{IV}^5)$ is square-integrable against the invariant Bergman measure, with the reproducing-kernel (Cauchy-Szegő / Shilov-boundary) structure fixing boundary behavior, so the spectrum is discrete and bounded below. Hence the gap is nonzero, and it is nonzero *because the domain is bounded*: decompactification ($R \rightarrow \infty$) sends $\lambda_1(R) \rightarrow 0$, killing the gap. *This compactness-is-the-gap-mechanism is the physical content, and it is not in question.* The gap has no free dimensionless parameter beyond the ruler — the geometry fixes everything up to the one dimensionful scale R (a Planck-unit, the same single input GR takes as G and QM as $\hbar$).

The bare- S^4 number “4” is real but is *projected out* on the physical boundary (Cal, K1709). On the *bare* four-sphere the Laplacian tower is $\lambda_k = k(k+3) = \{0, 4, 10, 18, \dots\}$ with degeneracies 1, 5, 14, 30, 55 — a genuine S^4 spectrum (not S^6 ; there is no S^6 in BST), whose first level ($k=1, \lambda=4$) has degeneracy exactly $n_C = 5$. But the physical boundary is not bare S^4 — it is $(S^4 \times S^1)/\mathbb{Z}_2$, and the $\mathbb{Z}_2$ identification couples the spatial S^4 level k to the time-circle mode m and admits only $k+m$ even. The $\lambda = 4$ mode is ($k=1, m=0$), for which $k+m = 1$ is odd — so it is projected out. The surviving lowest $\mathbb{Z}_2$ -even modes are R -dependent (e.g. $4/R^2$ or $4 + 1/R^2$, depending on which even (k,m) is lowest). So “ $\lambda_1 = 4$ ” is *not the physical gap*; the bare- S^4 computation was a valid intermediate that the real boundary removes.

Energy vs. Casimir, resolved (Grace $\oplus$ Elie, §570). The ν -independent number 4 is the spatial S^4 vector eigenvalue; the ν/Δ piece is time energy, not mass (the $SO(2)$ /time-circle conformal weight). These are *different objects on different circles* — the spatial Laplacian and the time-conformal weight must not be summed into one integer as if they were a single Casimir.

The honest value: three geometry-fixed opens (K1709). The physical gap on $(S^4 \times S^1)/\mathbb{Z}_2$ awaits three quantities, each fixed by the geometry (no dimensionless knob), none yet pinned: 1. the conformal weight Δ of the lowest gauge excitation (the time-circle / $SO(2)$ contribution); 2. the $\mathbb{Z}_2$ coupling — which surviving $k+m$ -even mode is the lowest; 3. the radius R (= the ruler) — which decides whether $4/R^2$ or $4 + 1/R^2$ wins.

These are genuine open physics, not a stalled identity. (*Use Heat_Trace’s own three-scale language: $\lambda_1, c_2 = 11$, and $C_2 = 6$ are each a distinct object — do not collapse them.*)

Why the old “6” and “7” were never the gap. “6” was the conformal reading — the bare 4 plus a +2 shift — which coincides with $C_2 = n_C + 1$ = the Bergman-rep Casimir (the identity Cal flagged, K1707; Guard 3), i.e. the conformal weight equated with its own definition, not an independent gap. “7” is the genus $g = n_C + 2$, a different invariant. Neither is λ_1 on the real boundary.

reading	value	status
bare- S^4 Laplacian first level, $k(k+3)$	4	real, but $\mathbb{Z}_2$ -projected-out — not the physical gap (K1709)
conformal reading ($4 + \rho$ -shift 2)	6	$= C_2 = n_C + 1$ — the identity (a time weight, not a mass)
Bergman-rep Casimir, $n+1$	6	$\equiv$ the identity (same definition)
genus / embedding invariant, g	7	a different invariant, not the gap
physical gap on $(S^4 \times S^1)/\mathbb{Z}_2$	awaits $\Delta + \mathbb{Z}_2$ -coupling + R	genuine open (K1709)

The glueball is a separate valid prediction, but out of the room for the gap (K1708/K1709). The glueball $1720 \text{ MeV} = (11/6) \cdot 938$ is a real forward prediction (Section 7) built on C_2^{int} (the labeled integer primary), not on λ_1 (the boundary Laplacian gap). It stays as a prediction, but it cannot corroborate the gap value — the same GATE-KOIDE discipline that keeps $Q = 2/3$ out of the Koide pin. The gap is pinned from the geometry alone.

Tier: gap EXISTENCE = SOLID (bounded $\Rightarrow$ gap, real physics; no free dimensionless parameter beyond the ruler). Gap VALUE = genuine open — three geometry-fixed unknowns (Δ , the $\mathbb{Z}_2$ coupling, R), the bare- S^4 “4” projected out. Floor = EXISTENCE (not ≥ 4). *Guard 1 holds: a gap of a boundary Laplacian on D_{IV}^5 , not asserted to be the $\mathbb{R}^4$ YM Hamiltonian’s gap.*

2a. A candidate floor-lift, if the mass/energy split holds (§570 — CANDIDATE, not banked)

The floor stays *existence*. But §570 (energy $\neq$ mass) suggests the floor may honestly lift, and we record it clearly tiered as a candidate, not banked.

The reading. On $(S^4 \times S^1)/\mathbb{Z}_2$ a mode (k, m) has spatial content k (the S^4 level) and time content m (the S^1 mode). By §570 the pure-time modes $(0, m)$ are *energies, not masses* — a state at rest in space but excited in time is not a massive particle. So the mass gap is the lightest mode that carries spatial structure ($k \geq 1$). And here the $\mathbb{Z}_2$ does something physical rather than merely subtractive:

The $\mathbb{Z}_2$ forbids a pure rest-mass (candidate BST statement about mass). A pure rest-mass would be a spatial-only mode ($k \geq 1, m = 0$). But $k = 1$ with $m = 0$ has $k + m = 1$, odd — forbidden. To be $\mathbb{Z}_2$ -even, a $k = 1$ spatial state must carry at least one unit of time, $m = 1$. So *on our boundary there is no purely-spatial massive state: every massive state carries one time-winding*. The projection that removed “4” is, read

forward, the statement that mass comes bundled with a minimal unit of time.

The candidate value. The lightest spatial-carrying $\mathbb{Z}_2$ -even mode is then $(k, m) = (1, 1)$, with energy

$$\Delta_{\text{mass}} = (\text{spatial } 4) + (\text{one time unit } 1/R^2) = 4 + 1/R^2 > 4 (= 5 \text{ at } R = 1).$$

So *if* the mass/energy separation holds on the real boundary, the mass gap returns above 4 with correct meaning — the pure-time modes $(0, m)$ are set aside as energies, and $(1,1)$ is the lightest genuine mass. This would lift the floor from existence to > 4 and identify the lowest $\mathbb{Z}_2$ -even *mass*-mode (opens ii partially), still carrying the ruler R (open iii).

Tier: CANDIDATE (not banked). This rides entirely on Grace $\oplus$ Elie confirming (a) the §570 mass/energy separation *on the real boundary* (that $(0,m)$ are energies, $(1,1)$ is the lightest mass), and (b) that “mass gap > 4 ” returns with the correct physical meaning. Until then the banked floor remains existence; the $(1,1) = 4 + 1/R^2$ reading is a *tiered candidate upgrade*, not a claim.

3. W4 dissolved: the gauge group is derived (Paper B)

A referee objection — “you treat only $SU(3)$ ” — is answered by showing G is not a free input. Paper B establishes D_{IV}^5 as the unique irreducible Hermitian symmetric domain meeting prior, dimension-innocent criteria, forcing $\dim_C = 5$ and $N_c = 3$ (the dimension-free output of the proved T1829). $N_c = 3$ is the T1829 output (a value-recurrence, Cal #335); the $su(3)$ /color readings of “3” are *downstream of the open color identification*, not independent derivations. So $SU(3)$ is substrate-derived. The whole dynamical content cascades from one physical input — three colors fix $\text{rank} = 2$ ($N_c = \text{rank}^2 - 1$), and rank generates $\{n_C, C_2, g\}$ (T2491). The “all G ” wall is dissolved.

4. W1 folded: the D_{IV}^5 spectral theory supplies the OS data on the 4D conformal boundary

BST does not construct 4D YM from scratch on $\mathbb{R}^4$. The OS reconstruction theorem builds a QFT from a data set, and the D_{IV}^5 spectral theory carries it — on the domain’s 4D conformal boundary, which exists because D_{IV}^5 carries the 4D conformal group $SO(4,2) \subset SO(5,2)$ (the same edge where light and electromagnetism live):

OS datum	D_{IV}^5 realization	tier
rigorous Hilbert space	the Hardy space $H^2(D_{IV}^5)$	SOLID

OS datum	D_IV ⁵ realization	tier
Hamiltonian with a gap	Laplacian on the real boundary $(S^4 \times S^1)/\mathbb{Z}_2$; gap nonzero (bounded $\Rightarrow$ gap), no dimensionless knob beyond R; value awaits $\Delta + \mathbb{Z}_2$ -coupling + R (bare- S^4 “4” $\mathbb{Z}_2$ -projected-out, K1709)	SOLID (existence); value open
reflection positivity	tube-type (Cayley $\rightarrow$ $T(\Omega)$); Θ = cone involution; OS-RP (free level) = Cauchy–Szegő RKHS-positivity	SOLID (free level)
4D conformal/Poincaré	$SO(4,2) \subset SO(5,2)$, UV-conformal/IR-gapped	SOLID-structural
clustering / unique vacuum	from the spectral gap $\Delta > 0$ (nonzero, geometry-fixed)	SOLID

So a rigorous gapped 4D theory follows from classical harmonic analysis; the open from-scratch construction is sidestepped, not solved. The single non-free-level piece is the interacting upgrade of reflection positivity — which is the residual (Section 6). Tier: SOLID at free level; interacting-RP = the residual.

5. Net-compatibility via Bisognano–Wichmann (SOLID-CONDITIONAL)

The HS (Hardy–Szegő) isometry intertwines the bulk and boundary operator nets: HS is $SO(5,2)$ -equivariant, and both the bulk Rehren net and the boundary net are modular reconstructions of the *same* $SO(5,2)$ positive-energy representation via Bisognano–Wichmann (BGL) / Borchers. So locality/causality transfer across HS. Tier: SOLID-CONDITIONAL on the BGL hypothesis-checks. (*This is the modular-geometry rung of the reconstruction stack named in Section 9.*)

6. The one residual: interacting Yang–Mills, or generalized-free? (= “is the Rehren edge literally YM?”)

The prize is a single question, and F1056 and v0.3 name it two ways that are one residual:

Is the OS/Rehren-reconstructed gapped 4D boundary theory genuinely interacting $SU(3)$ Yang–Mills, or a generalized-free gapped theory?

The two namings coincide: - v0.3’s naming (interacting vs generalized-free): a generalized-free field also satisfies the OS axioms and also has a gap — so the

residual is to show the reconstructed theory is genuinely *interacting*. - F1056's naming (Rehren-edge = literally-YM): Rehren's algebraic holography is a rigorous bulk $\leftrightarrow$ boundary net correspondence, but its *dual is not automatically the physical CFT* — so the residual is to show the Rehren edge theory is *literally* SU(3) YM, not a cousin.

These are the same open piece. “Genuinely interacting” and “literally the physical YM net” are one condition, viewed from the OS side and the Rehren side respectively.

A trap we avoid. “Non-commutative $\implies$ interacting” is false — a free field also has a non-commutative CCR algebra. Interaction means nonzero connected n-point functions ($n \geq 3$). The YM self-interaction is the vertex $[A,A]$, nonzero precisely because the gauge algebra is non-abelian. So

interacting $\iff$ the gauge algebra is genuinely non-abelian $\mathfrak{su}(3) \iff$ the cross-channel glueball spectrum matches SU(3) (Section 7).

Two pieces of evidence toward the interacting side: - The bulk-color algebra closes as non-abelian $\mathfrak{su}(3)$ (bilinear Schwinger realization; Elie 4301). The substrate identification (#418) is in progress. - The spectrum is non-additive and computed (Section 7). A generalized-free theory has an additive multi-particle spectrum and no genuine bound-state J^{PC} tower. The computed ladder is non-additive and channel-structured — evidence of binding. This is *evidence toward* interacting, not a closure of it.

7. The cross-channel glueball spectrum — named-open, materially advanced

In v0.1 this section was *named-open at structural tier*. Since v0.3 it is computed and materially advanced — still named-open (the honest tier, after Cal's trim of the v0.2 over-claim “derived / landed interacting”).

7.1 Why there are exactly four channels (operator-algebraic closure)

The gauge field F is a rank-2 antisymmetric tensor operator on $H^2(D_{IV}^5)$. Its bilinears $F \otimes F$ decompose into exactly four irreducible J^{PC} components, each of which must have eigenstates for the substrate's operator algebra to close:

channel	operator	what it commits
0^{++}	$\text{Tr}(F^2)$	energy density (the action)
0^{-+}	$\text{Tr}(F \tilde{F})$	topology (Pontryagin density)
2^{++}	$\text{Tr}(F^{\mu\rho} F^{\nu}_{\rho})$ traceless	curvature (the stress tensor)
1^{+-}	$\text{Tr}(F[D,F])$	derivative structure

The multiplicity is forced by the tensor structure — the answer to “why four channels?” Same architectural principle as nuclear magic numbers, different driver

(operator-algebra closure for the bosonic tensor vs Pauli for fermions). The odd-balls (0^{+-} , 1^{-+} , 2^{+-}) are forbidden at two gluons — clean unmixable channels.

7.2 The masses: one verified number sets the ladder

The glueball mass is the eigenvalue of the linear conformal Hamiltonian (the $SO(2)$ dilatation) on the holomorphic discrete series on $H^2(D_{IV}^5)$, diagonal in the K-type basis by Schur:

$m \propto E = \lambda_0 + (\text{energy step})$, $\lambda_0 = \text{genus}(D_{IV}^5) = n_C = 5$ (verified from the multiplicity formula $p = (r-1)a + b + 2 = 5$).

Energy step = $SO(5)$ harmonic degree = spin J ; the parity-odd (Hodge-dual) sector carries the half-canonical twist $n_C/2$. With one dimensionful anchor (seat = $\pi^5 \cdot m_e = 156.4$ MeV; $m(0^{++}) = c_2 \cdot \text{seat} = 1720$ MeV):

channel	step	ratio	substrate form	BST (MeV)	lattice (MeV)	dev	tier
0^{++}	0	1	—	1720	1730	−0.6%	anchor
2^{++}	$J = 2$	$7/5$	g/n_C	2408	2400	+0.3%	BLIND (genus + spin)
0^{-+}	$n_C/2$	$3/2$	N_c/rank	2580	2590	−0.4%	consistent (twist value-checked)
1^{+-}	$1 + n_C/2$	$17/10$	—	2924	2940	−0.5%	consistent (twist value-checked)

Only the $2^{++} = g/n_C$ leg is fully blind (genus + spin only, nothing read from the data, lands because $g = n_C + \text{rank}$). The 0^{-+} and 1^{+-} use the half-canonical twist $n_C/2$, which is rep-motivated but value-checked against the data — so they are consistent within lattice error, not blind predictions. (Quantization note: the *ratios* live on the rung ladder $\lambda_0 = n_C$; the absolute scale comes only from anchoring 0^{++} at seat 11.)

7.3 The radial direction is linear, by Schur

The scalar holomorphic discrete series is one irreducible representation; by Schur the Casimir is constant across its K-types, so it cannot distinguish radial levels. The first radial scalar excitation $0^{++*} = (1,1)$ sits at the linear conformal energy $E = \lambda_0 + 2 = 7$, degenerate with 2^{++} (2408 MeV; lattice ~ 2670 , within quenched excited-glueball error). Consequence stated honestly: the spin-linear / radial-quadratic factorization is a clean negative — both directions are linear within the irrep.

7.4 Experimental and decay-side notes

The experimental scalar-glueball candidate $f_0(1710) \approx 1704$ MeV is consistent with BST's 0^{++} at seat 11 (1720 MeV) — the expected lightest-glueball match (not a unique identification; the scalar sector mixes with $q\bar{q}$). The 0^{-+} coupling f_G is the substrate-computable Bergman mode norm (a Gindikin-Gamma ratio; 0^{++} kernel $= 60 = C_2 \cdot n_C \cdot \text{rank}$); with the established $\chi_{\text{top}}^{1/4} = 180$ MeV the WV identity gives $f_G(0^{-+}) \approx 12.6$ MeV. Decay-side: mixing (glueball $\leftrightarrow q\bar{q}$ overlap on the shared Hardy space), not a “dump.”

7.5 Honest disposition of Section 7

The cross-channel spectrum is computed (four channels; one blind leg $2^{++} = g/n_C$; the others consistent within lattice error; multiplicity explained; radial direction settled by Schur; lightest state consistent with $f_0(1710)$). This materially advances the interacting reading — a generalized-free theory produces no such bound-state J^{PC} tower. It does not close the interacting-vs-free residual (Section 6). That — equivalently, the *formal interacting-RP upgrade / proving the Rehren edge is literally YM* — remains the named-open residual (Section 9).

8. The $so(7)$ unification (LEAD-STRENGTHENED)

Color and the Yang–Mills spectrum live in one algebra: $su(3) \subset g_2 \subset so(7)$, and $so(7)$ is the compact dual isometry of Q^5 — the same $so(7)$ whose Casimir spectrum is the glueball tower of Section 2; $g = 7$ is the $so(7)$ vector $= g_2$ fundamental $= 3 \oplus \bar{3} \oplus 1$. Color is not a geometric isometry of the noncompact domain ($su(3)$ not-subset-of $so(5,2)$) but is a geometric subalgebra of the compact-dual $so(7)$; on the domain side it is the operator algebra on H^2 . The discrete-series spectrum of this $so(7)$ has the primaries $\{N_c, n_C, C_2, g\}$ as its lowest half-Casimirs (T2490). Tier: LEAD-STRENGTHENED — the algebraic picture is whole; the bilinear-Schwinger realization on H^2 is in progress (#418).

9. Honest scope: the conditional, and the reconstruction stack for the one residual

We do not claim to have proved the Yang–Mills mass gap. The honest statement:

Given the reconstruction stack executes to rigor — including the interacting upgrade that identifies the edge theory as literally $SU(3)$ YM — the physical Yang–Mills theory exists as a rigorous 4D QFT with a nonzero, geometry-fixed mass gap (no free dimensionless parameter beyond the ruler; the integer value set by the boundary data $\Delta + \mathbb{Z}_2$ -coupling $+ R$ on $(S^4 \times S^1)/\mathbb{Z}_2$, K1709).

The reconstruction stack (the honest W4 machinery, named — F1056/K1705/T1271):

Hua-1963 boundary values $\rightarrow$ Bisognano–Wichmann-1975 / Borchers-2000 modular geometry $\rightarrow$ Rehren-2000 holographic net $\rightarrow$ Osterwalder–Schrader reconstruction $\rightarrow$ cluster decomposition.

Rehren’s algebraic holography is a *theorem* (rigorous, not the approximate AdS/CFT), so the residual is well-posed, not hand-waving. But it has two named, unfinished parts: (1) run the correspondence on D_{IV}^5 (build the bulk net and obtain the 4D boundary net); (2) identify the reconstructed boundary theory with the physical $SU(3)$ YM — the known subtlety that a Rehren/OS dual is not *automatically* the physical CFT. Part (2) is the genuine open piece, and it is *the same* as the interacting-vs-free residual (Section 6).

What is SOLID independent of the conditions: the gap exists and is nonzero (bounded $\Rightarrow$ gap) with no free dimensionless parameter beyond the ruler R (Section 2 — the integer value awaits three geometry-fixed opens: the conformal weight Δ , the $\mathbb{Z}_2$ coupling, and R ; the bare- S^4 “4” is $\mathbb{Z}_2$ -projected-out, K1709); the substrate-derivation of $SU(3)$ (Paper B); the OS *data* (Section 4); the $so(7)$ unification (Section 8). The glueball 1720 MeV at 0.6% is a real, separate forward prediction (built on C_2^{int} , not λ_1) but out of the room for the value-pin (K1708/K1709). What is materially advanced but still named-open: the cross-channel spectrum (Section 7) — real evidence on the interacting side that does not by itself close the residual. The single flat-space residual is the interacting-upgrade / literally-YM identification. This is a sharpening of the Clay problem — *not* a claim of its solution, and (after K1709) *not* a claim of a derived integer gap value; the gap value is honest open physics, floored at existence.

Explanatory summary (Guard 1 + Guard 2 held). BST *reproduces the scale and channel-structure of the Yang–Mills mass gap on a bounded domain*: the gap $\Delta = C_2 = 6$ is forced by representation theory (Guard 1: a domain operator, not the $\mathbb{R}^4$ Hamiltonian), its pure-glue reading matches the glueball at 0.6% (Guard 2: not the proton), and the flat- $\mathbb{R}^4$ construction is carried as the large named open — never a proof.

10. Open items and falsifiers

- ★ The mass-gap operator on the real boundary (the live task, K1709): define the operator on $(S^4 \times S^1)/\mathbb{Z}_2$ (not bare S^4), and pin the three geometry-fixed opens — (i) the conformal weight Δ of the lowest gauge excitation, (ii) which surviving $\mathbb{Z}_2$ -even ($k+m$ even) mode is lowest, (iii) the radius R (deciding $4/R^2$ vs $4+1/R^2$). Glueball out of the room (GATE-KOIDE).
- The one flat-space residual (Section 6/9): the interacting-upgrade / proving the Rehren-OS edge theory is literally $SU(3)$ YM.
- #418 substrate identification: the bulk-color Toeplitz octet *is* the bilinear-Schwinger $su(3)$ (algebra closure verified; identification in progress).
- D_A dynamical-gap check (Elie): does the fluctuated Dirac D_A^2 carry a positive gap dynamically, and at what value on the *real* boundary (state the space first; do not pre-fill “ $C_2 = 6$ ” — that is the projected-out/identity reading)? Report “the D_{IV}^5 gap,” target the glueball, not the proton.
- Falsifier: the cross-channel glueball spectrum is a forward prediction — 0^{++} degenerate with 2^{++} at 2408 MeV is testable; if the cross-channel ratios fail, the interacting reading fails (the gap existence-floor and Paper B survive).

Count HOLDS 4 of 26. SU(3) scope; glueball masses are predictions, not SM parameter reductions. Paper A v0.8 (K1709 + §570 candidate): the gap exists (bounded $\Rightarrow$ gap, no free dimensionless parameter beyond the ruler) — this floor is rock-solid. On the real boundary $(S^4 \times S^1)/\mathbb{Z}_2$ the $\mathbb{Z}_2$ admits only $k+m$ even, so the bare- S^4 “4” ($k=1, m=0$) is projected out (Cal); the honest value awaits three geometry-fixed opens: (i) the (adjoint) conformal weight Δ , (ii) which $\mathbb{Z}_2$ -even mode is lowest, (iii) the radius R . Candidate floor-lift (Section 2a, tiered — not banked): by §570 the pure-time modes $(0,m)$ are *energies not masses*, so the *mass gap* is the lightest spatial-carrying $\mathbb{Z}_2$ -even mode $(1,1) = 4 + 1/R^2 > 4$ ($= 5$ at $R=1$) — the $\mathbb{Z}_2$ forbids a pure rest-mass, so every massive state carries one unit of time; this returns “gap > 4 ” with correct meaning *if* Grace $\oplus$ Elie confirm the mass/energy split on the real boundary. The old “6” was the conformal reading = the identity $C_2 = n_C + 1$; “7” = genus. The glueball 1720 is a separate valid prediction (on C_2^{int} , not λ_1) — kept, out of the room as gap-evidence. Section 0 keeps the three C_2 guards + the glueball exemption. Package intro HELD until Grace $\oplus$ Elie define the operator on the real boundary, test the §570 lift, and pin $\Delta + \mathbb{Z}_2$ -coupling + R . NOT a derived value (the $(1,1)$ lift is a tiered candidate), NOT “spectrum closed,” NOT “YM proved.” INTERNAL. Nothing pushed; CP existence-only.

Draft v0.8 (K1709 + §570 candidate). Spine: Cal’s $\mathbb{Z}_2$ projection catch (floor = existence); Grace $\oplus$ Elie’s energy-vs-Casimir resolution (§570) + the real-boundary operator definition + the §570 mass/energy-split test (the live tasks); Lyra’s three-opens framing + the §570 candidate floor-lift (mass gap = lightest spatial mode $(1,1) = 4 + 1/R^2$; $\mathbb{Z}_2$ forbids a pure rest-mass — every massive state carries one time unit), clearly tiered CANDIDATE; the so(7) unification; Paper B (gauge-group derivation); Keeper K1709. Existence banked; the $(1,1)$ lift is a candidate. Companion to Paper B v0.6.

— Lyra, Wed 2026-08-19 (date-verified). Paper A v0.8: floor = EXISTENCE (banked); §570 candidate floor-lift added (Section 2a, tiered) — mass gap = lightest spatial-carrying $\mathbb{Z}_2$ -even mode $(1,1) = 4 + 1/R^2 > 4$, because the $\mathbb{Z}_2$ forbids a pure rest-mass (every massive state carries one unit of time); glueball separate + out of the room; package intro HELD.